means rotational power is transmitted to main shaft by 1-2-4-7-S_1.

Second gear

For obtaining the second gear, by actuating the gear shift lever, the synchronizer S_2 is moved towards right to engage the gear 8 in the output main shaft. Now the rotational power is transmitted to the main shaft through the synchronizer S_2. Now the speed of the main shaft is higher than the speed of the first gear. The power transmission from the clutch shaft to the output shaft is through the gears 1-2-3-8-S_2.

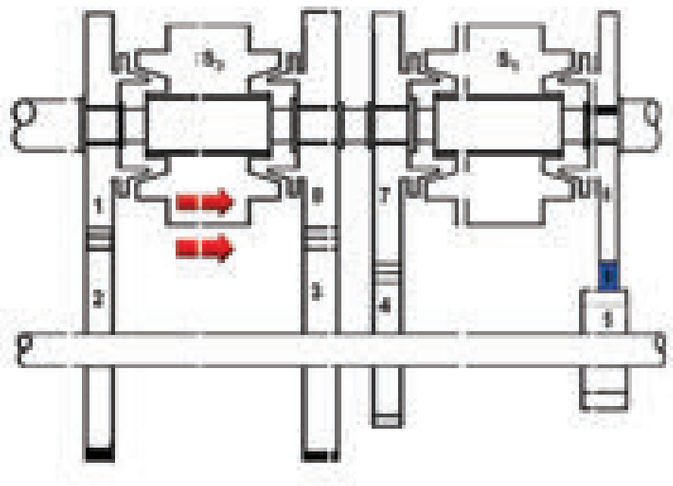


Figure 3.5.3.3 Second gear condition

Third gear

To achieve the third gear the synchronizer S_2 is moved left to engage with the clutch shaft gear directly. Now the output shaft rotates at the speed same as the clutch shaft speed. The rotational power is transmitted to the main shaft directly through the gear 1-S_2.

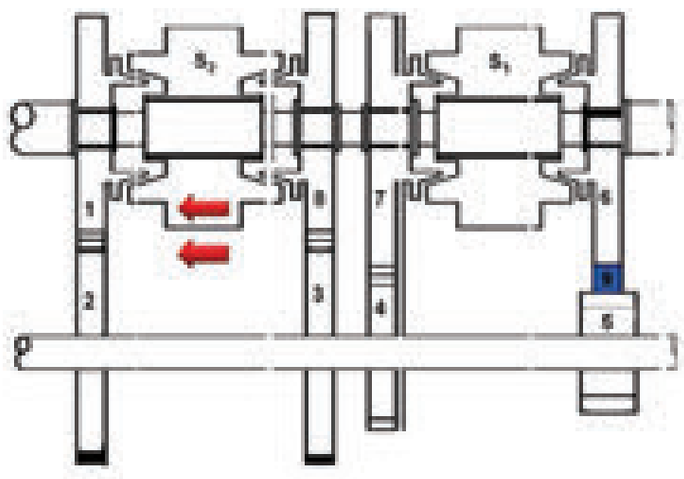


Figure 3.5.3.4 Third gear condition

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